

Sensation

CT

Troubleshooting Guide

PHS (Standard and Bariatric)

Troubleshooting Guide PHS

This document is valid for:
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Handling of Patient Table

All couch operating functions (horizontal and vertical) can be executed from the gantry operating panels in the gantry doors as well as from the GPC board in the right gantry stand (Gantry Panel Control / D311).

There are two operating panels integrated in the gantry front door (left & right) and now two additional control panels at the gantry rear door that enable all couch movements.



Fig. 1: Gantry operating panel (front)



Fig. 2: Gantry operating panel (rear)

The couch also has foot switches (2 at each corner of the couch base) that allow vertical up/down movement. These switches perform the same functions as the corresponding buttons on the gantry operating panels, and they are likewise controlled by the GPC board (D311). (Fig. 4 / p. 6)

Service switches at GPC board (D311)

The GPC-board (D311) is located in the right gantry stand and allows the identical functions via onboard push buttons (with LEDs) as the gantry control panels.

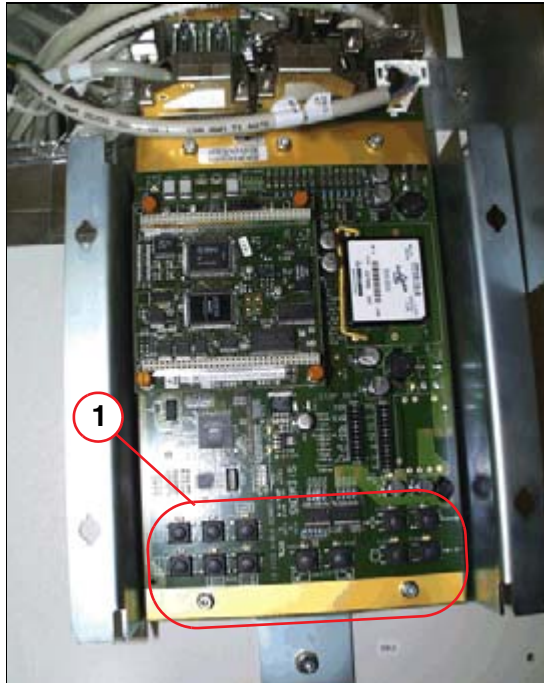


Fig. 3: GPC board (right gantry stand)

Pos. 1 Service switches

Switch ID	Function description
S01	Gantry tilt positive/forward (viewed from top of gantry)
S02	Gantry tilt negative/backward (viewed from top of gantry)
S03	Horizontal table movement positive / FEED OUT(slow speed)
S41	Horizontal table movement positive / FEED OUT(fast speed)
S10	Horizontal table movement negative / FEED IN (slow speed)
S42	Horizontal table movement negative / FEED IN (fast speed)
S12	Vertical table movement positive / DOWN
S13	Vertical table movement negative / UP
S20	Move patient from outer light marker to scan slice position
S21	Withdraws table for patient loading or unloading
S23	Resets horizontal table position to 0
S30	Light marker switch on/off

Service switches at the couch

There are service switches (S703 / S701 / S702) at the couch base (right front corner) that enable vertical movement in service mode.

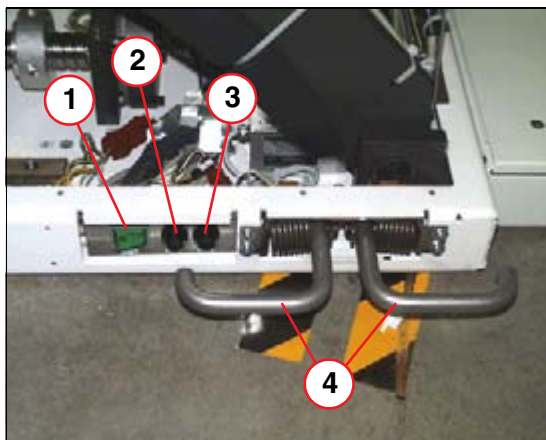


Fig. 4: Service switches “vertical”

- Pos. 1 S703 / Service ON / OFF
- Pos. 2 S701 / UP
- Pos. 3 S702 / DOWN
- Pos. 4 Footswitches UP / DOWN

Switch ID	Function description
S703	Service ON / OFF switch for vertical movement
S701	Vertical movement UP (service)
S702	Vertical movement DOWN (service)

Location of Patient Table Components

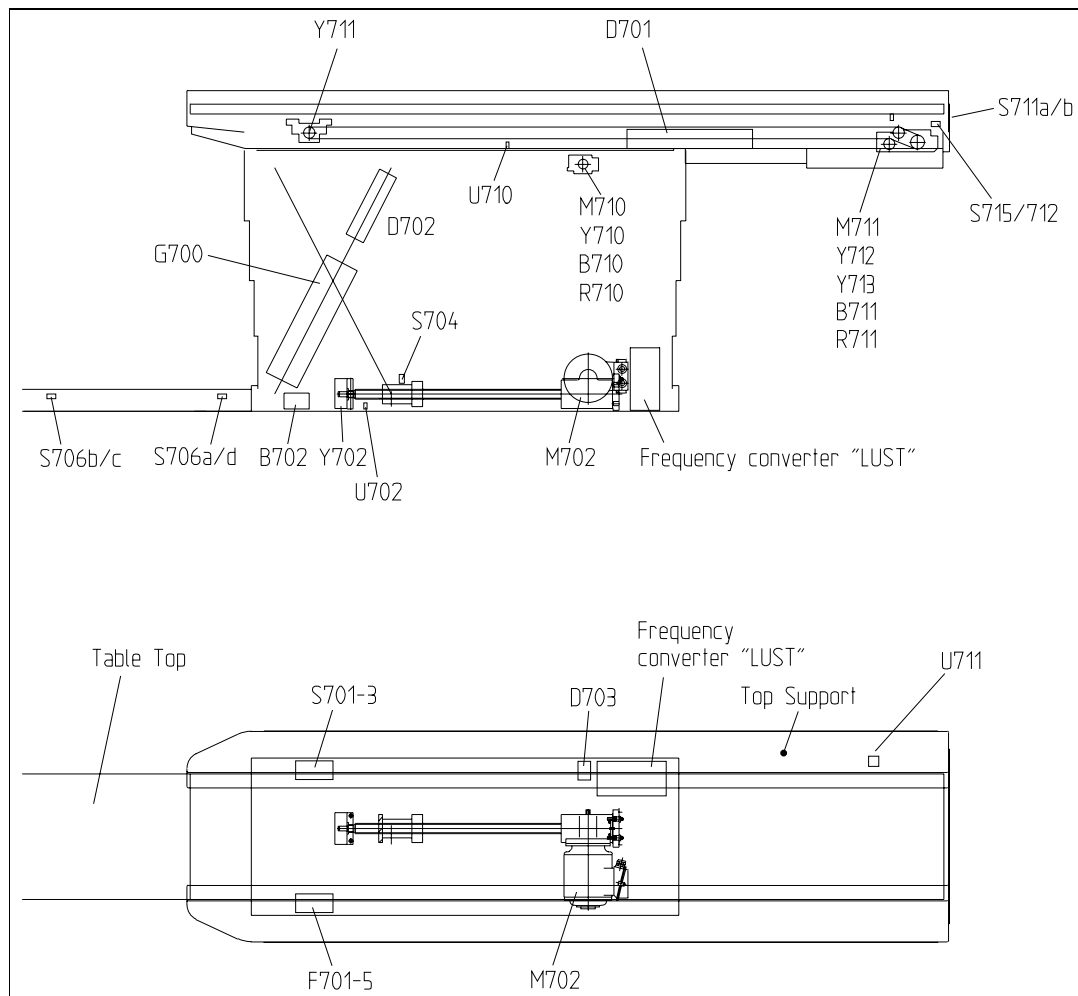


Fig. 5: Overview - Couch components

This chapter describes all fuses and LED's at the couch as well as on PHS-relevant components (e.g. PDS in PDC cabinet, GPC board at right gantry stand)

The tables should allow a quick overview and judgement of the status and the condition of various functions.

LED's

There are LED's on serveral electronic components in the couch that allow a quick overview and check of their status and condition.

Overview of all PHS-relevant LED's

The following table lists all PHS-relevant LED's, their location, name, normal status and function.

Tab. 1 PHS-relevant LED's

Location / Component	LED# (colour)	Status (normal)	Function
PTH / D701	V38 (green)	ON	24V for Top support
	V39 (green)	ON	24V for Tabletop
	V45 (green)	ON	5V VCC-Voltage OK
	V9 (green)[on MCB2]	Flashing	CAN-Communication active ???
PTV / D702	V9 (green)	ON	+5V OK
	V10 (green)	ON	-12V OK
	V11 (green)	ON	+12V OK
Frequency converter (LUST)	H1 (red)	OFF	Warning/Error indication if ON or flashing
	H2 (yellow)	ON or Flashing	Ready for Operation / Operation active
	H3 (green)	ON	Power to frequency converter is ON
CAN-Module(at frequency converter)	H4 (red)	OFF	Error indication if ON
	H5 (green)	ON	Ready for Operation & Communication
AC/DC converter G700	U1 (green)	ON	+5 VDC Output-Voltage OK
	U2 (green)	ON	+12 VDC Output-Voltage OK
	U3 (green)	ON	+12 VDC Output-Voltage OK
	U4 (green)	ON	+24 VDC Output-Voltage OK
	"Power ON"	ON	+230 VAC Input-Voltage OK

Frequency converter

The different possible combinations of the LED's H1/H2/H3 at the frequency converter indicate a certain component status. Please refer to the table below for more details.

Tab. 2 LED-combinations at frequency converter

LED H1(red)	LED H2 (yellow)	LED H3 (green)	Component status:
-	-	ON	Power (supply voltage) is ON
OFF	ON	-	Ready for operation
OFF	Flashing	-	Operation active
ON	ON or Flash- ing	-	Warning
Flashing (error code)	OFF	-	Error

In case of an error situation at the frequency converter (LUST) a certain LED flashing sequence is displayed with the red H1 LED. Refer to the table below for the different error codes and suggested service actions.

Tab. 3 Flash code description of LED H1 (red) at frequency converter

Flash code LED H1(red)	Error status	Suggested service action:
1x	CPU error	Switch power to PHS OFF /ON. Replace frequency converter if error still active.
2x	Low voltage or power interrupt	Check power distribution.Switch power to PHS OFF /ON.
3x	Over current	Check vertical drive motor and cabeling
4x	Over voltage	Check power, switch power to PHS OFF/ON
5x	Motor protection	Check vertical drive motor
6x	Frequency converter protection	Switch power to PHS OFF /ON. Replace frequency converter if error still active.
7x	Motor temperature too high	Check vertical drive motor
8x	Over temperature	Switch power to PHS OFF /ON. Replace frequency converter if error still active.

Fuses and circuit breakers

The following table gives an overview of all PHS-relevant fuses and circuit breakers and lists their location, name, value and function.

Tab. 4 PHS-relevant fuses and circuit breakers

Location / Component	Fuse / Circuit breaker #	Value	Function
PDC cabinet	F8	3 x 13A	Main circuit breaker for gantry (PSS) and PHS power. Phase L2 for PHS power.
	F15	6A	220V for solid state relays (D703) and frequency converter (LUST). Signal: L2_PHS1
	F16	2A	220V for AC/DC converter (G700) and ECG power supply (if available). Signal: L2_PHS2
PTH board / D701	F1	1AT	Supply voltage for trolley (not used)
	F2	1,5AT	5V for VCC
	F3	5AT	24V for top support
	F4	5AT	24V for tabletop
Base of couch (left side)	F701	3A	5V for PTH/D701 (from AC-DC converter/G700)
	F702	3A	5V for PTV/D702 (from AC-DC converter/G700)
	F703	3A	12V for PTV/D702 (from AC-DC converter/G700)
	F704	3A	24V for PTH/D701 (from AC-DC converter/G700) Signal: Supp_24V
	F705	3A	24V for PTH/D701 (from AC-DC converter/G700) Signal: Table_24V

Functions

PT HORIZONTAL = Patient Table Horizontal Control Unit

The sub-assembly PT HORIZONTAL (D701 plus MCB2 D2) controls, regulates, and monitors all horizontal movements of the tabletop and the tabletop support. All horizontal functions can be performed via GPC board D311 and all gantry operating panels.

NOTE

The CAN-Bus is the communication medium for controlling and performing table functions.

Measurement points & signals

PT Horizontal board (D701)

Signal name	Measur- ment point	Description
MP_SUPP_SW	X224	Reference switch for top support travel. The top support is in the zero position (maximum distance to gantry) and the table can be lowered.
MP_TABLE_SW	X225	Reference switch for tabletop travel. Table can be lowered (starting position / zero position) for table travel.
MP_TABLE_CT	X228	Pulses from the horizontal movement of the incremental encoder / tabletop. 1 mm in table travel corresponds to approximately 2000 encoder pulses or 200 HTP pulses.
MP_TABLE_DIR	X229	Measurement point for horizontal tabletop travel. 'H' means in/towards the gantry. 'L' means out/away from the gantry.
MP_HTP	X227	HTP pulses from the PLD-Logic. The HTP pulses are also counted on the ACU (D513) in the C-Box.

Tests

FDA Test

During the FDA Test, the table is moved five times into the gantry and out again for a total of 300 mm in 10 mm increments. The tabletop positioning is verified by comparing the pre- and post-movement accuracy of the table positions.

Starting the FDA Test

- Select: **Options/Local Service**.
- Select: **Test Tools/Controller/PTH**.
- Select: **FDA Test**.
- Select: **GO** to start the test.
- Follow the instructions on the screen.

Involved PHS components

The following components are involved in the vertical couch movement:

- AC-DC converter (G700)
- PTV board (D702)
- Frequency converter (LUST)
- Solid state relais (D703)
- Spindle drive (M1)
- Spindle brake (Y702)
- Vertical position sensor (B702)
- Safety switch S704

AC-DC converter (G700)

The AC-DC converter provides the necessary supply voltages (5V, 2 x 12V) to the PTV board.

D702 / PT Vertical board

The PT Vertical board (D702 + MCB2 D2) controls the vertical movement and position of the couch. All vertical functions can be performed also via the GPC-board D311 and the gantry operating panels.

The CAN-Bus is the communication medium for controlling and performing all table functions.

D703 / Solid state relais

The D703 is a solid state relais that works as a stop relais. It is controlled by the system and can interrupt the power to the frequency converter in emergency situation (e.g. stop circuit active).

Frequency converter (LUST)

The frequency converter controls the vertical movement of the spindle drive. The communication between the D702 and the frequency converter (CAN-module) is performed via CAN-bus.

Spindle drive M1

The spindle drive contains the drive motor M1 and the spindle. It is responsible for the mechanical movement of the table top.

Spindle brake Y702

The spindle brake Y702 is attached to the spindle drive. It is responsible for a secure mechanical stopping and blocking of the vertical movement.

Vertical position sensor (B702)

The vertical position sensor is a tow-rope potentiometer that is responsible for the display of the actual vertical position.

Safety switch S704

The safety switch S704 has a safety function that deactivates the vertical drive in case of a blocked table top when lowering the table.

Foot switches

There are four pairs of foot switches at the base of the couch (one set on each corner) that allow up/down movement. The footswitches are directly controlled by the GPC board (D311) in the right gantry stand

Troubleshooting sequences

The following trouble shooting sequences are described in a logical order and should guide the service engineer in a fast and effective way to the source of the problem.

Check of error messages

In case of malfunctions during vertical movement the PTV controller generates certain PTV-relevant error messages.

- Check the eventlog for any PTV-relevant error messages.
-

Check of vertical movement in Service mode

- Switch the couch into vertical service mode with switch S703 (right side at base of couch).
 - ⇒ NOTE: It takes a view seconds before the frequency converter is ready for operation (LED H2 must be ON).
- Raise the couch via switches S701 (UP) and then lower the couch via switch S702 (DOWN).
 - ⇒ If movement is possible all mechanical components (spindle drive and brakes) and also the frequency converter and solid state relais must be OK.
 - ⇒ If movement is not possible go to section "Check of power distribution"
- Switch OFF service switch S703.

Check of power distribution to PHS

- Check the circuit breakers F8, F15 and F16 in the PDC cabinet.
 - ⇒ All circuit breakers must be ON.
- Check the service switch S2 at the PDC.
 - ⇒ Switch S2 must be OFF (control lamp H2 must be OFF).
- Check the LEDs at the AC-DC converter (G700) in the couch.
 - ⇒ All 5 LED's must be ON.
 - ⇒ If an LED is OFF check the according fuse at the table base (left side).
 - ⇒ If all LEDs are OFF check the 220V AC power distribution to the AC-DC converter (signals: L_PHS2 and N_PHS2)
- Check the LED's (V9/V10/V11) at the PTV board.
 - ⇒ All 3 LEDs must be ON.
- Check the LED H3 at the frequency converter
 - ⇒ LED H3 must be ON
 - ⇒ If LED H3 is OFF check the input voltage at the solid state relais (D703)

Check the vertical position sensor (B702)

- Slightly pull on the measuring thread of the vertical position sensor and monitor the display at the gantry front door.
 - ⇒ The vertical position value at the gantry display must change.
 - ⇒ If vertical position value does not change check the cable connection of the sensor at the PTV-board (D702.X5).

Tests

Test Name	Test Description
Test "Motor Brake"	The vertical travel of the table is stopped by both the motor brake and the spindle brake. At a defined height (vertical position 445 mm) the spindle brake is released so that the motor brake alone must stop the vertical travel of the table. If the table is lowered more than a defined distance, the spindle brake is actuated and the system reverts to the STOP mode.
Test "Spindle Brake"	The vertical travel of the table is stopped by both the motorbrake and the spindle brake. At a defined height (vertical position 445 mm) the motor brake is released so that the spindle brake must stop the vertical travel of the table. If the table is lowered more than a defined distance, the motor brake is actuated and the system reverts to the STOP mode.

Test "Motor Brake"

- Select: **Options/Local Service**
- Select: **Test Tools/Controller/PTV**
- Select: Test **Motor Brake**
- Select: **GO** to start the test
- Follow the instructions on the screen.

Test "Spindle Brake"

- Select: **Options/Local Service.**
- Select: **Test Tools/Controller/PTV**
- Select: Test **Spindle Brake**
- Select: **GO** to start the test
- Follow the instructions on the screen.

Chapter	Change	Reason
FMP	SOMATOM Sensation 40 and the Information (Standard and Bariatric) was added	136660